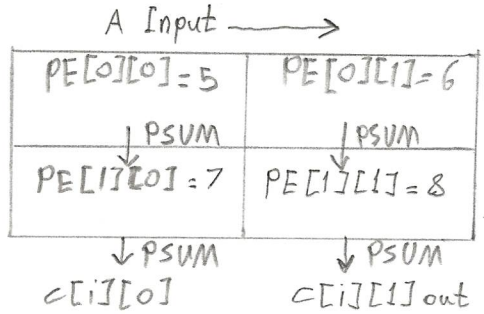
Codefest 5

1-



2-

Cycle	Input to row 0	Input to row 1	PE[0][0] PSUM	PE[0][1] PSUM	PE[1][0] PSUM	PE[1][1] PSUM	Output C
1	1	/	1X5=5	/	/	/	/
2	3	2	3X5=15	1X6=6	5+2X7 = 19	/	C[0][0] =19
3	/	4	/	3X6=18	15+4X7 =43	6+2X8 =22	C[1][0]=43 C[0][1]=22
4	/	/	/	/	/	18+4X8 =50	C[1][1] =50

C=[[19,22],[43,50]]

3- a) For 2 X 2 matrix, multiplying the inner dimension result in 2X2X2=8 MACs

b)

A Value	Used by	# of uses	# of reuses
A[0][0]=1	C[0][0], C[0][1]	2	1
A[0][1]=2	C[0][0], C[0][1]	2	1
A[1][0]=3	C[1][0], C[1][1]	2	1
A[1][1]=4	C[1][0], C[1][1]	2	1

B Value	Used by	# of uses	# of reuse
B[0][0]=5	C[0][0], C[1][0]	2	1
B[0][1]=6	C[0][1], C[1][1]	2	1
B[1][0]=7	C[0][0], C[1][0]	2	1
B[1][1]=8	C[0][1], C[1][1]	2	1

c)

Data	off chip access
A input reads	4
B input reads	4
C output writes	4
total	12

d) The output C[i][j] would stay fixed in the PE